

Exact Computations on the $K3 \times T^2$ Reading of the Cooper s_7/s_{10} Families, and What the Simulation Cross-Checks Do and Do Not Show

Xavier Callens

SocrateAI Scientific Agora Collaboration

Stream 2 — Selection & Geometry

(with the simulator stream's ledger, as reported by that stream)

2026-10-10 (preprint; not reviewed)

What is not claimed

This is a report on exact and numerical *mathematics*. It contains no physical claim and no observable. The physical reading of “ $K3 \times T^2$ ” as a compactification, and any statement about a dark sector, are **Tier C — blocked physics** and are not tested, validated or supported by anything here. No torus of any physical model is identified with an elliptic curve E or E' . Nothing ranks the s_7 family against the s_{10} family. No Kodaira fibre type is attached to any point of either family.

Abstract

We collect the numerical results the programme has obtained for the Shioda–Inose reading of $K3 \times T^2$ and say, for each, what it is worth. Every table is generated from a certificate. The transcendental lattice of $E \times E'$ for a cyclic n -isogeny is computed exactly in $\wedge^2 \mathbb{Z}^4$ and is isometric to each family's certified lattice, for $n = 7$ and $n = 10$. At the three Picard-number-20 points of s_7 the lattice of $E \times E$ for the computed CM order equals the certified row; that agreement is forced by theory and is not evidence. A level-consistency gate between the lattice and modular legs holds for both families. At $z = 1/27$ the Mordell–Weil section demanded by the M_7 polarization is constructed by the Kumar–Kuwata isogeny graph, reconstructed as a rational function from 52 exact specializations (an over-determined fit with held-out points), and read: $\bar{P} \cdot \bar{O} = 5$, contact 0 with the A_1 fibre, height 14, equal to the lattice prediction. We then report, with counts and without re-deriving them, what the DualScaleSimulator stream's own cross-check ledger records, and what its level experiments have and have not established. The physical theory is not validated, and we state what a validation would require.

Contents

1	What is computed, and the tier of each result	1
2	The lattice of $E \times E'$	1
3	Level consistency	2
4	The section at $z = 1/27$	3
5	The simulation side: what the cross-check ledger records	4
6	What cannot be concluded, and what would be needed	5

1 What is computed, and the tier of each result

Result	Tier	Source of the number
$T(E \times E') \cong U \oplus \langle 2n \rangle$, $n = 7, 10$	B (exact computation)	K3xT2_READING_S.json
$T(E \times E)$ at the three s_7 points of $\rho = 20$	B, forced	same
Level consistency, lattice $n =$ modular n	B (hard gate)	T3_LEVEL_CONSISTENCY.json
The section at $z = 1/27$: $\bar{P} \cdot \bar{O}$, contact, height	B (exact; construction L)	TW2_SECTION_DESCENT.json
Counts of the simulator stream's cross-check ledger	mirrored, not verified here	refs/simulator_k3t2_ledger_counts_v3.json
Identification of the T^2 of a compactification with E	C, not made	—
Any dark-sector mass, coupling or exclusion	C, blocked	—

Table 1: Scope of this report. B: derived, not measured. L: literature. C: blocked physics.

2 The lattice of $E \times E'$

Let $E \times E' = \mathbb{C}^2/\mathbb{Z}^4$. Then $H^2(E \times E', \mathbb{Z}) \cong \wedge^2 \mathbb{Z}^4$ with the pairing $\langle u, w \rangle =$ the coefficient of $e_1 \wedge e_2 \wedge e_3 \wedge e_4$ in $u \wedge w$, and a two-dimensional subtorus spanned by v_1, v_2 has class $v_1 \wedge v_2$. For curves without complex multiplication (an assumption, stated and not computed), the Néron–Severi lattice is spanned by the two fibres and the graph of the isogeny. The checker verifies that this span has signature $(1, 2)$ and is saturated, computes its orthogonal complement as an integral basis, and finds an explicit unimodular basis change onto the certified lattice. Controls that must fail do: a wrong level, a non-cyclic isogeny (caught by saturation), a reversed orientation (caught by the signature), a tampered lattice, a wrong CM order, and a j -invariant that is not a class-number-one value.

Family	n	NS sig.	saturated	$T(E \times E')$ (Gram), sig.	$\cong$ certified T
cooper_s7	7	(1, 2)	yes	$[0, 0, -1; 0, 14, 0; -1, 0, 0]$, (2, 1)	yes
cooper_s10	10	(1, 2)	yes	$[0, 0, -1; 0, 20, 0; -1, 0, 0]$, (2, 1)	yes
s_7 point	model j	CM disc.	$T(E \times E)$	certified T	agree
1/27	16581375	−28	$[2, 0; 0, 14]$	$[2, 0; 0, 14]$	yes (forced)
−1	−3375	−7	$[2, 1; 1, 4]$	$[2, 1; 1, 4]$	yes (forced)
∞	0	−3	$[2, 1; 1, 2]$	$[2, 1; 1, 2]$	yes (forced)

Table 2: Top: the computed $T(E \times E')$ against each family's certified T . Bottom: $T(E \times E)$ at the s_7 points of Picard number 20, with the CM order identified by computation from the model's j . The bottom agreement is forced (an elliptic point of the modular curve is a CM point); it is consistency, not corroboration.

The Shioda–Inose isometry $T(X) \cong T(E \times E')$ for an Inose surface X is cited (Tier L — literature, Kumar–Kuwata Remark 2.2), not recomputed. For s_7 the family member is identified with an Inose surface through the explicit M_7 -polarized model; for s_{10} , which has no explicit model here, the reading holds at the lattice level only. The rows of s_{10} at $\rho = 20$ are not covered.

3 Level consistency

Family	n from lattice	n from modular side	Verdict
cooper_s7	7	7	T3_AGREE($n=7$)
cooper_s10	10	10	T3_AGREE($n=10$)

Table 3: The level n read from the lattice certificate against the level read from the modular side (an eta-quotient Hauptmodul with Newman’s conditions and an Atkin–Lehner check). Both legs start from the same operator, so agreement is not an independent measurement. The gate has real negative controls (four other order-three entries fail its modular leg) and its power against a genuine two-leg candidate that disagrees is unestablished.

4 The section at $z = 1/27$

At the three $\rho = 20$ points of s_7 the two curves of the Inose model become isomorphic and the count of table 4 follows from exact reduction of the model and the height formula $h(P) = 2\chi + 2\bar{P} \cdot \bar{O} - \sum \text{contr}$.

s_7 member	$ \text{disc } T $	ρ	extra roots	MW rank	$h(P)$	$\bar{P} \cdot \bar{O}$
generic	14	19	—	1	14	5
$z = 1/27$	28	20	A_1	1	14	5
$z = -1$	7	20	A_1	1	7/2	0
$z = \infty$	3	20	A_2	0	—	no section

Table 4: The generic member and the three $\rho = 20$ points of s_7 . The point $z = \infty$ has Mordell–Weil rank 0: there is no section there.

For $z = 1/27$ the section is the one attached to the degree-7 endomorphism $\sqrt{-7}$ of E . We constructed it, rather than inferring it from the lattice, by the following exact procedure.

1. The Vélú isogeny $\varphi: E_1 \rightarrow E_2$ of degree 7 is written over $\mathbb{Q}$ (kernel cubic, codomain), and $\varphi_y = \varphi'_x$ is checked exactly against the curve equation.
2. The cubic C_t of Kumar–Kuwata (3.1) is reduced, through $s = x_2 - t^2 x_1$, to a quartic with a rational point, and mapped to a Weierstrass model E_t by Mordell’s formulas. The map is verified by identity on the actual divisor points, and $j(E_t)$ by the quartic’s own invariants.
3. The divisors $D^\pm$ ($\varphi_y(x_1) = \pm t^3$, nine points) are summed by Riemann–Roch: the function in $L(10O)$ vanishing on $D^\pm$ has one further zero R , and $Q^\pm = -R$. The section is $P = Q^+ - Q^-$.
4. E_t is matched to the simple normal form $Y^2 = X^3 - 3\alpha X + t_s + t_s^{-1} - 2\beta$ ($\alpha = (J_1 J_2)^{1/3}$, $\beta = J - 1$, $t_s = t^6/343$, the constant $343 = \sqrt{\Delta_{E_1}/\Delta_{E_2}}$ computed); the j -match and the twist constant are asserted at every specialization.
5. $X'(t_s)$ is reconstructed as a rational function from exact specializations and *read*.

Headline result

The explicit section at $z = 1/27$ meets O five times, once inside the A_1 fibre (hence on its identity component) and four times at the roots of an irreducible quartic. Its height is $14 = 2 \cdot 7$, the value the lattice arithmetic and the Kumar–Kuwata theorem both give. The two E_8 fibres carry no contact. **Tier B — derived, not measured**.

Quantity	Value
fit points / unknowns / held out	44 / 26 / 8
nullspace dimension (1 = unique)	1
$\deg N$, $\deg D^2$ of $X' = N/D^2$	14, 10
held-out points agree	yes
$\bar{P} \cdot \bar{O}$ (read)	5
contact with the A_1 fibre, contr (read)	0
height $h = 2\chi + 2\bar{P} \cdot \bar{O} - \text{contr}$ (read)	14
earlier steps and literature, read from their certificates	step 0: $h = 14$; step 2a: $\bar{P} \cdot \bar{O} = 5$, $\text{contr} = 0$; Kumar–Kuwata: $h = 14$
discriminant orders (model; fibration certificate)	$\{10, 10, 2, 1, 1\}$; $\{10, 10, 2, 1, 1\}$
cross-check agrees	yes

Table 5: The reconstruction and what is read from it. The fit is over-determined and agrees at held-out points. The cross-check row compares with the earlier lattice steps and with the fibration certificate, all read from their own files.

Remark 4.1. A control found a robustness fact worth recording: $\bar{P} \cdot \bar{O}$ equals $(\deg N - 4)/2$ whatever the denominator, because poles that leave the finite line reappear at infinity. The *count* is fixed by the degree of the numerator; the pole *locations* carry the contact information. Doubling the section does not fit the same shape, as it should (its height is four times larger).

What this does not establish: that the section generates the whole Mordell–Weil group (no index computation was made; the lattice side and the literature both give height 14, which is the natural reading, not a proof here); anything at $z = -1$ (where the lattice count has $\bar{P} \cdot \bar{O} = 0$) or at $z = \infty$; or any physical statement.

5 The simulation side: what the cross-check ledger records

The DualScaleSimulator stream maintains a ledger (its `k3t2_rigidity_v3` audit) comparing, for 332 sealed targets of the Lean library, values it computed in a separately written pipeline against the values stated in Lean. We report its *counts*, recounted from its row-level data by a script that refuses on any mismatch, pinned to the source file’s hash. We did not re-derive its rows: nothing in table 6 is verified by this stream, and the source’s own header places its computations at tier B at best and its physical identifications at tiers L and C.

Count in the simulator stream’s ledger	Value
sealed targets compared	332
AGREE	68
SAME_DECLARED_INPUT (downgraded)	3
NOT_COMPUTED	261
DISAGREE	0
rows with two or more independent routes	6
declared inputs (of which typed from memory)	80 (20)
rigidity entries: confirmed / disputed / unreviewed	3 / 2 / 28
chain links consistent; independent corroborations	8 (all); 3
items its authors could not do	30

Table 6: The simulator stream’s ledger, mirrored (source commit and hash pinned in `refs/`). “Rigid” there means that a selecting condition which never mentions the target value was scanned for one parameter; it is not a statement about the universe.

Two features of the table matter for how it may be used. First, most targets are not computed: the ledger is a map of what remains, not a verdict on the library. Second, a fifth of the inputs it declares were typed from memory; the ledger says so itself, and a number resting on such an input is only as good as the memory. Of the rigidity entries, three are confirmed, two are disputed between its two reviewers and the rest are self-reported and unreviewed.

Its level experiments. Quoting the simulator stream’s own note (`audit/K3_SELECTION.md`, not recomputed here): the implemented model sits at level 12 because a constant is hard-coded, so level 12 is fitted, not forced; under the potential actually integrated no observable depends on the level, so no level can be falsified; and a modular potential that does read the level exists but is opt-in and not adopted. Its estimate is that, in one of its two modes, about three quarters of the apparent level dependence of that potential is a stretched axis and not structure (over 14 random points). The test that would show whether the level reaches a reported observable through the dynamics (its item L5) ran only partially and is neither met nor refuted.

6 What cannot be concluded, and what would be needed

The physical theory is not validated. There is no worked matching from the geometric data to a four-dimensional effective theory, the flux and tadpole route is blocked until a threefold base is specified, and the pre-registered prediction carries no derived quantity. A validation would require, in order: a worked matching; a quantitative prediction that differs from Λ CDM; a pre-registered pin; and survival against public data. None of the first link exists. The mathematics reported here is groundwork for that chain and says nothing about whether it can be completed.

We did not compute $\text{tr } G + \text{tr } G^{-1}$ on the tori of the CM points. The Lean-proved dual-scale bound is a statement about all positive-definite metrics; evaluating it at special points would need a typed Kähler modulus, is not invariant under change of basis without an orbit minimum, would identify the torus of an elliptic curve with a physical torus, and would order the two families. None of these is available.

7 Corrections recorded in band

Corrections recorded in band

1. The reading of “ $K3 \times T^2$ ” used here was first described as the programme’s choice before it had been ruled. It was ruled on 2026-10-08 (decision D29’, taken on the decision authority’s behalf under explicit delegation) and is a convention for selection criteria only.
2. The first reconstruction of the section used a transcription of the Weierstrass form from a garbled line of the source. The exact j -match test refused it, and the simple normal form of the same source was used instead; the constant $t_s = t^6/343$ was then calibrated on three points and is asserted at all fifty-two.
3. A first sensitivity control for the read-off passed for the wrong reason (removing the quartic moves its poles to infinity, leaving the total unchanged). It was replaced by one that adds a genuine pole.
4. A note elsewhere in the programme said the host lacked a numerical library and that two test modules were ignored. That was a property of the system interpreter; under the project’s own environment the full suite runs. Corrected 2026-10-10.
5. One continuous-integration run of the simulator failed once on a tolerance of 10^{-12} in an ODE-solve comparison and passed on re-run of the same commit; the tolerance is hardware-sensitive and was not changed.

8 Reproducibility

Every number in tables 2 to 6 is read from a certificate or a pinned mirror by `scripts/render_paper_tables.py`, whose `-check` mode fails on drift and whose controls include tampered inputs for each table. The checkers and their controls:

- `checkers/check_K3xT2_reading_S.py` (9 controls);
- `checkers/check_T3_level_consistency.py` (25 controls);
- `checkers/check_TW2_section_descent.py` (14 controls);
- `scripts/extract_simulator_ledger_counts.py` (recount; 8 controls).

Decision records: `briefs/TO_DECISIONS_2026_10_07_STREAM2_DELEGATED.md` (D29’–D31’) and `briefs/STREAM2_K3xT2_DUAL_SCALE_SCOPING_2026_10_10.md`.

Provenance

Generated-by: Claude (Fable 5.1 and Sonnet 5.5), Stream 2 | Verified-by: `render_paper_tables.py` `-check` and its controls; each cited checker’s own controls | Reviewed-by: N (preprint; not reviewed).

References

- [1] A. Kumar and M. Kuwata, “Elliptic $K3$ surfaces associated with the product of two elliptic curves,” *arXiv:1409.2931*.

- [2] M. Schütt and T. Shioda, “Elliptic surfaces,” *arXiv:0907.0298*.
- [3] T. Takatsu, “On the geometry of singular $K3$ surfaces with discriminant 3, 4 and 7,” *arXiv:1903.03054*.
- [4] I. Dolgachev, “Mirror symmetry for lattice polarized $K3$ surfaces,” *arXiv:alg-geom/9502005*.
- [5] X. Callens and the SocrateAI Scientific Agora Collaboration, “The selected $K3$ surfaces of the Cooper s_7/s_{10} families: identification, the $K3 \times T^2$ reading, and the twisted-Weierstrass route,” Stream 2 note, 2026-10-08 (papers/stream2_k3_identification_twisted_route_2026_10_08.pdf).